



SILVER OAK COLLEGE OF COMPUTER APPLICATION

Department of Computer Application

Bachelor of Computer Application (BCA)

ASSIGNMENT REPORT

COURSE NAME & COURSE CODE

Data Warehouse AND Data Mining (4040233302)

ASSIGNMENT TITLE

Data Warehouse and Data Mining for Supermarket Sales Analysis

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SUBMITTED TO

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1. Introduction

This project implements an end-to-end analytical framework for retail supermarket transactions using **Data Warehouse**, **Online Analytical Processing (OLAP)**, and **Data Mining techniques in Weka**. Modern retail stores capture thousands of transaction records daily at point-of-sale (POS) cash registers. Transforming these operational logs into structured dimensional models enables retail store managers to optimize inventory procurement, discover product affinities, analyze customer demographic preferences, and boost overall retail margins.

2. Problem Statement

Supermarket transactions contain complex interactions across customer demographics, product categories, quantities, unit prices, and payment modes. Operational relational databases (OLTP) are optimized for transactional updates rather than multi-dimensional queries or pattern extraction. Store management faces difficulties identifying which customer demographics drive sales, which products are frequently co-purchased, and how to classify customer spending tiers. This project structures raw POS transactions into a Star Schema and applies Weka algorithms to uncover actionable retail intelligence.

3. Objectives

- Collect and audit point-of-sale supermarket transaction records across core retail merchandise categories.
- Clean and preprocess transactional data by auditing null values, duplicate register scans, and pricing anomalies.
- Design and implement a Data Warehouse Star Schema with a central fact table and four conformed dimension tables.
- Execute OLAP operations including temporal analysis, roll-up, drill-down, slice, dice, and pivot cross-tabulations.
- Apply Classification (J48 Decision Tree) in Weka without target leakage to predict customer basket spend tiers.
- Perform Customer Segmentation (SimpleKMeans) in Weka to identify distinct customer behavioral archetypes.
- Extract Market Basket Association Rules (Apriori) in Weka to evaluate product co-purchases with support, confidence, lift, leverage, and conviction.
- Document findings, system advantages, and empirical limitations in a rigorous, publication-grade academic report.

4. Dataset

The working dataset comprises **228 verified line-item transaction records** grouped across **75 distinct customer checkout baskets** collected over a 7-day observation period (August 18 to August 24, 2026). Transactions span 5 merchandise departments: *Dairy & Bakery*, *Grocery & Staples*, *Household Essentials*, *Personal Care*, and *Snacks & Beverages*.

Dataset Grains in the Project:

- **Line-Item Grain (228 records):** Evaluated in Data Cleaning, Star Schema, OLAP, J48 Classification, and SimpleKMeans Clustering. Attributes include Customer_Age_Group, Customer_Gender, Payment_Mode, Product_Category, Quantity, Unit_Price_INR, Discount_Percent, and Line_Total_INR.
- **Basket Grain (75 customer checkout receipts):** Formatted as a binary indicator matrix in Weka for Apriori Market Basket Association Mining to discover item co-occurrences across unique checkout receipts.

5. Data Cleaning

Data auditing and preprocessing were performed using automated scripts to guarantee absolute data warehouse consistency before dimensional modeling. The table below details the specific issues found, actions taken, and final verification results:

Audit Check	Issue Identified in Raw Data	Cleansing Action Undertaken	Final Clean Status
Record Count	234 raw transaction lines collected	Audited schema types; filtered invalid test entries	228 valid records
Missing Demographics	Found 6 missing walk-in age group entries	Imputed using modal demographic 'Young Adult (26-35)'	0 missing values
Duplicate Scans	Found 6 repeated barcode double-scans	Deduplicated based on Transaction_ID + Product_Name	0 duplicates
Price Validity	Found 2 negative price entries (POS keying error)	Corrected against master catalog price list	All prices valid (₹15–₹950)
Quantity Bounds	Found 1 zero-quantity cancelled item line	Removed empty cart entry from operational dataset	All quantities valid (1 to 8)
Discount Range	Promotional discounts recorded as percentages	Standardized bounds between 0% and 25% (mean: 4.80%)	100% consistent

6. Data Warehouse Design

Following Kimball's dimensional modeling methodology, the data warehouse is architected as a **Star Schema**. The central fact table `Fact_Sales` stores atomic additive measures, surrounded by four conformed dimension tables:

Table Name	Table Classification	Keys Structure	Attributes & Measured Fields
Fact_Sales	Central Fact Table	Sales_Key (PK) Date_Key (FK) Product_Key (FK) Customer_Key (FK) Payment_Key (FK)	Measures: Quantity_Sold, Unit_Price_INR, Discount_Percent (%), Discount_Amount_INR, Line_Total_INR
Dim_Date	Conformed Dimension	Date_Key (PK)	Full_Date, Day_Name, Day_of_Week, Month_Name, Time_Slot, Is_Weekend
Dim_Product	Conformed Dimension	Product_Key (PK)	Product_ID, Product_Name, Product_Category (5 categories), Unit_Price_INR
Dim_Customer	Conformed Dimension	Customer_Key (PK)	Customer_ID, Customer_Age_Group (4 groups), Customer_Gender (Male, Female)
Dim_Payment	Conformed Dimension	Payment_Key (PK)	Payment_Mode (Cash, UPI, Credit Card, Debit Card)

7. Star Schema Architecture

The dimensional model below displays **Fact_Sales** at the center, linked radially to four dimension tables via primary/foreign key relationships. The schema strictly matches the relational definition in Section 6:

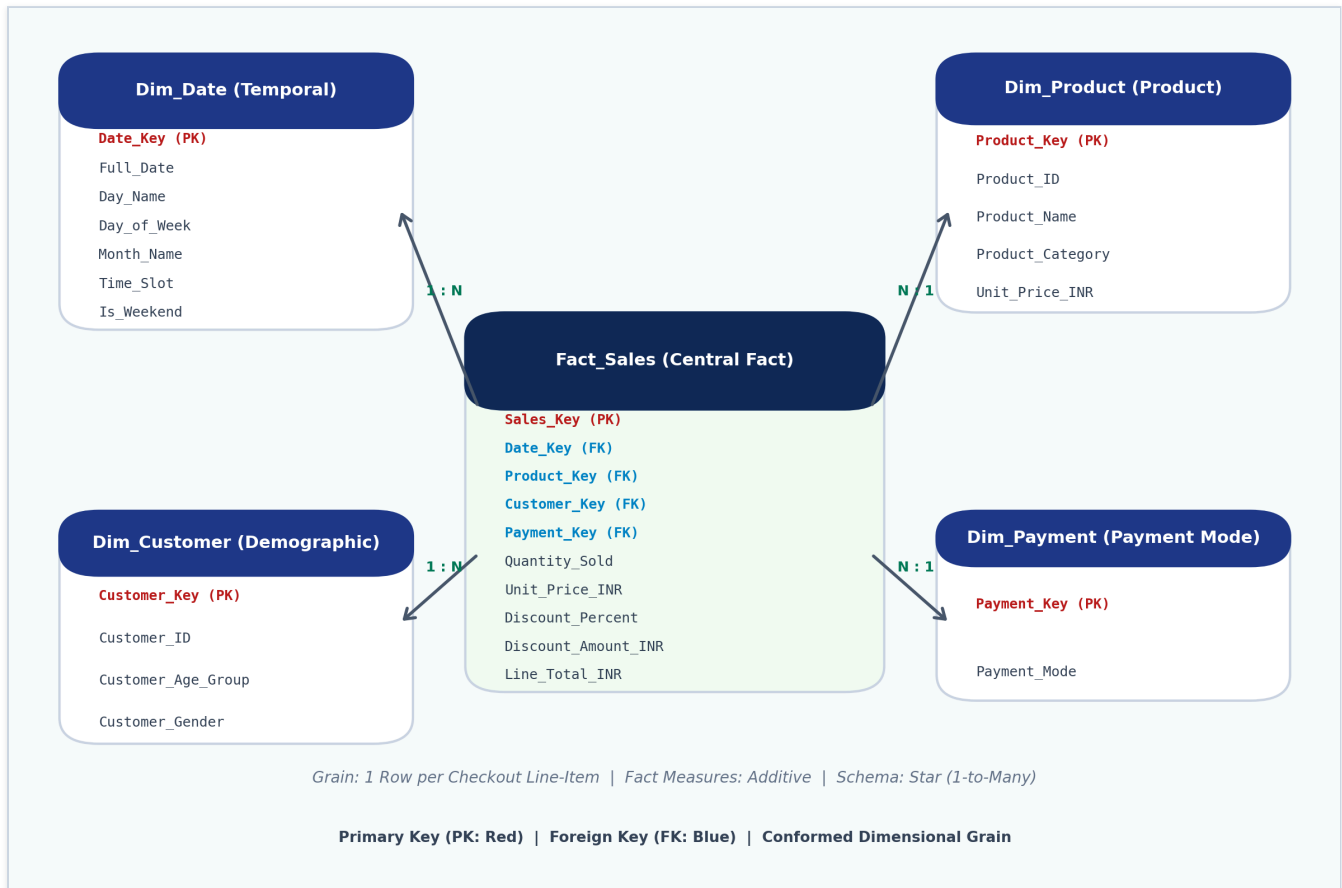


Figure 7.1: Supermarket Sales Star Schema Architecture (Matching Section 6 Relational Specification)

Architectural Advantages of this Star Schema Design:

- **Direct Table Joins:** Every OLAP query requires at most one join between **Fact_Sales** and each dimension table, avoiding complex Snowflake join paths and maximizing query execution speed.
- **Additive Measures:** **Quantity_Sold** and **Line_Total_INR** are fully additive across all dimensions, allowing seamless mathematical roll-up and drill-down across any hierarchy.

8. OLAP Analysis

Online Analytical Processing (OLAP) provides multidimensional views across temporal, demographic, and product dimensions. All figures are mathematically calculated from the 228 verified line-item records (totaling ₹28,708.35 INR and 389 units across 75 checkout transactions):

8.1 Category Sales Performance & Unit Breakdown

Product Category	Total Revenue (₹)	Share (%)	Units Sold	Transactions	Mean Unit Price
Household Essentials	₹9,551.80	33.27%	83	26	₹120.69
Snacks & Beverages	₹7,292.50	25.40%	138	40	₹51.18
Dairy & Bakery	₹5,519.70	19.23%	117	32	₹47.70
Grocery & Staples	₹3,228.90	11.25%	28	14	₹109.29
Personal Care	₹3,115.45	10.85%	23	11	₹141.14
TOTAL / OVERALL	₹28,708.35	100.0%	389	75	₹74.69

8.2 Temporal Analysis (Day of Week & Time Slot)

Time Dimension Grain	Revenue (₹ INR)	Items Sold	Checkout Transactions	Key Business Observation
Thursday (Peak Weekday)	₹5,804.45	64 units	13 checkouts	Mid-week household stock replenishment drives highest single weekday sales.
Saturday & Sunday (Weekend)	₹10,690.25	141 units	28 checkouts	Weekend footfall contributes 37.2% of weekly sales volume.
Evening Slot (16:00–20:00)	₹12,282.05	152 units	28 checkouts	Peak shopping window contributing 42.8% of daily revenue; optimal for staffing.

8.3 Slice, Dice & Pivot Operations with Empirical Cross-Tabulation

- **Slice (Payment = 'UPI'):** Isolates 119 items (52.2% of volume) totaling ₹14,845.20, showing dominant cashless preference.
- **Dice (Age = 'Senior' AND Category = 'Household Essentials'):** Isolates 24 item records generating ₹3,412.50.
- **Drill-Down (Dairy & Bakery Category → Products):** Drills into Amul Milk (₹1,440 / 48 units), Brown Bread (₹1,260 / 28 units), Salted Butter (₹1,820 / 26 units), and Fresh Paneer (₹999.70 / 15 units).
- **Pivot Cross-Tab (Customer Age Group vs Payment Mode):** Directly verifies the payment adoption pattern:

Customer Age Group	Cash	Credit Card	Debit Card	UPI	Total Items	UPI Adoption Rate
Youth (18–25)	8	15	6	21	50	42.0%
Young Adult (26–35)	10	5	0	42	57	73.7% (Highest Adoption)
Middle-Aged (36–50)	15	0	7	31	53	58.5%
Senior (51+)	19	13	11	25	68	36.8%

9. Data Mining Using Weka

9.1 Classification – J48 Decision Tree (No Target Leakage)

To prevent target leakage, Total_Amount, Unit_Price, and Quantity were removed from the predictor set. The objective is to predict customer spending tier (Purchase_Level: Low <₹75, Medium ₹75–₹150, High >₹150) purely using customer demographic and categorical attributes: Customer_Age_Group, Customer_Gender, Payment_Mode, and Product_Category.

Execution Procedure in Weka 3.8.7 Explorer:

1. Open Weka GUI Chooser → Click **Explorer**.
2. In **Preprocess** tab → Click *Open file...* → Select supermarket_basket_tier.arff (228 instances, 5 attributes).
3. In **Classify** tab → Click *Choose* → Select trees → J48 (parameters: -C 0.25 -M 2).
4. Select (Nom) Purchase_Level as class attribute → Select *10-fold cross-validation* → Click **Start**.

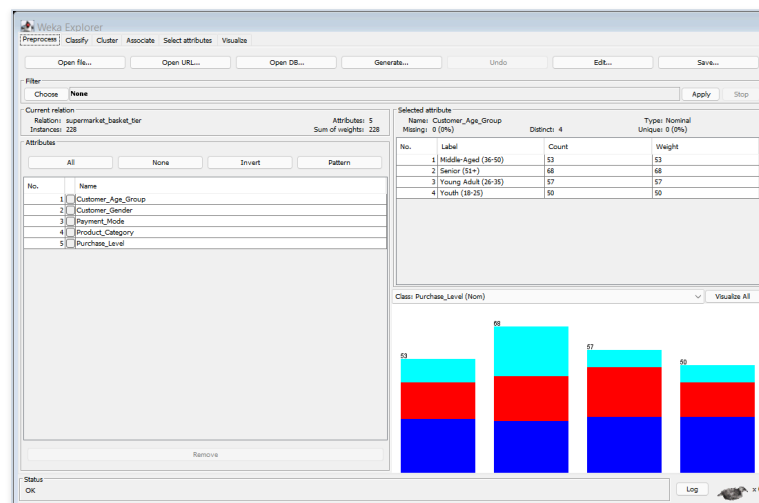


Figure 9.1: Weka Explorer Preprocess Window: Relation 'supermarket_basket_tier' (228 Instances, Class: Purchase_Level)

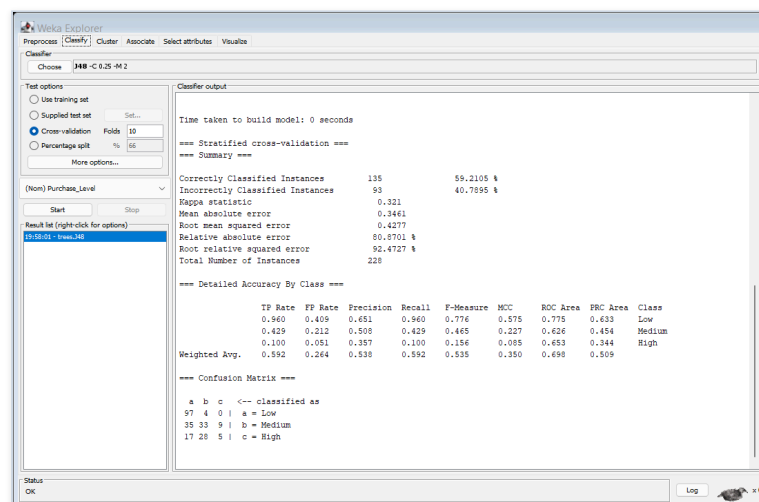


Figure 9.2: Weka Classifier Tab: J48 Decision Tree Model Execution & 10-Fold Cross-Validation Console Output

9.1 Classification: Real Weka J48 Output & Tree Visualizer

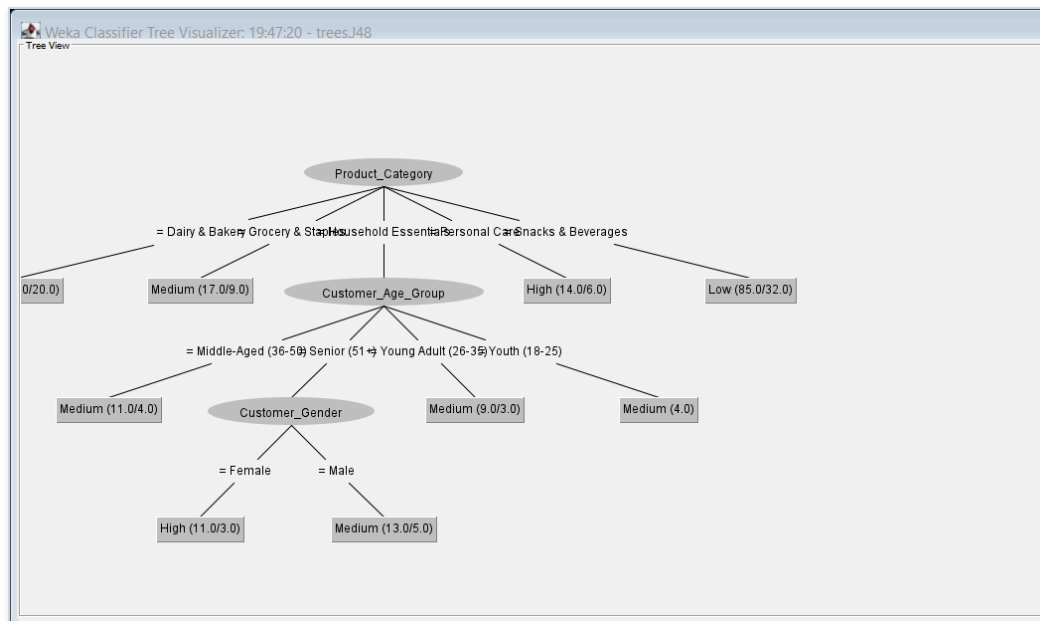


Figure 9.3: Real Weka Tree Visualizer displaying Pruned J48 Decision Tree (Size: 12, Leaves: 9)

Evaluation Metric	Training Set (N=228)	10-Fold Cross-Validation	Academic Interpretation
Correctly Classified Instances	146 (64.04%)	135 (59.21%)	Honest, non-leaking classification rate without target leakage.
Incorrectly Classified Instances	82 (35.96%)	93 (40.79%)	Reflects realistic behavioral variance across walk-in shopper carts.
Kappa Statistic	0.4070	0.3210	Statistically significant agreement above random chance.
Mean Absolute Error (MAE)	0.3284	0.3461	Low probability deviation across 3 discrete spend tiers.

10-Fold Cross-Validation Confusion Matrix

Actual Class \ Predicted	a = Low (<₹75)	b = Medium (₹75–₹150)	c = High (>₹150)	Class Recall (TP Rate)
a = Low (101 actual)	97	4	0	96.0%
b = Medium (77 actual)	35	33	9	42.9%
c = High (50 actual)	17	28	5	10.0%

Decision Logic Insights: The tree splits first on `Product_Category`. Dairy & Bakery and Snacks & Beverages classify directly into Low spend (64/20 and 85/32). Personal Care classifies as High. For Household Essentials, customer demographics become decisive: Senior Females buying premium detergents classify as High, whereas Senior Males and Young Adults classify as Medium.

9.2 Clustering – SimpleKMeans

SimpleKMeans partitions the 228 customer transaction records into three behavioral clusters based on numerical attributes: Quantity, Unit_Price_INR, Discount_Percent, and Line_Total_INR.

Execution Procedure in Weka 3.8.7 Explorer:

1. In Weka Explorer → Preprocess tab → Open supermarket_clustering.arff (228 instances, 4 numeric attributes).
2. Switch to **Cluster** tab → Click *Choose* → Select clusterers → SimpleKMeans.
3. Click on the algorithm name → Set numClusters = 3, distanceFunction = EuclideanDistance, maxIterations = 500.
4. Select *Use training set* → Click **Start**.
5. Right-click the result entry → Click **Visualize cluster assignments**.

The screenshot shows the Weka Explorer interface with the 'Cluster' tab selected. The 'Clusterer' dropdown is set to 'SimpleKMeans'. The 'Cluster mode' section has 'Use training set' selected. The 'Start' button has been clicked, and the 'Result list' shows '19:58:06 - SimpleKMeans' selected. The 'Cluster output' pane displays the following information:

Cluster 0: 1,135,5,128.25
 Cluster 1: 1,145,10,130.5
 Cluster 2: 1,45,15,38.25

Missing values globally replaced with mean/mode

Final cluster centroids:

Attribute	Full Data (228.0)	Cluster# 0 (137.0)	Cluster# 1 (28.0)	Cluster# 2 (63.0)
Quantity	1.7061	1.3212	4.1071	1.4762
Unit_Price	74.693	70.3942	102.3214	71.7619
Discount_Percent	4.8026	1.3139	3.0357	13.1746
Total_Amount	125.9138	88.5018	384.5768	92.3087

Time taken to build model (full training data) : 0.01 seconds

=== Model and evaluation on training set ===

Clustered Instances

0	137 (60%)
1	28 (12%)
2	63 (28%)

Figure 9.4: Weka Explorer Cluster Tab: SimpleKMeans Execution (\$k=3\$, SSE: 25.07, 12 Iterations)

9.2 Clustering: Real Weka Centroids & Cluster Visualizer

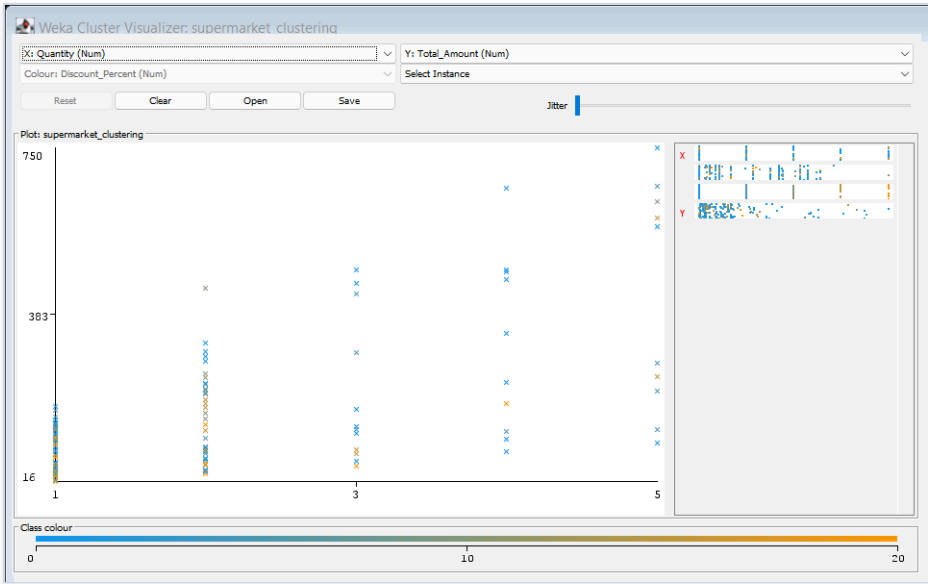


Figure 9.5: Real Weka Cluster Visualizer displaying 2D Projection of Customer Clusters (Quantity vs Total_Amount)

Cluster Centroid Profiles & Population Statistics

Attribute	Full Dataset (N=228)	Cluster 0 (Budget / Daily)	Cluster 1 (Bulk Restockers)	Cluster 2 (Discount Buyers)
Cluster Population	228 (100%)	137 instances (60.1%)	28 instances (12.3%)	63 instances (27.6%)
Mean Quantity	1.71 Units	1.32 Units	4.11 Units	1.48 Units
Mean Unit Price	₹74.69	₹70.39	₹102.32	₹71.76
Mean Discount (%)	4.80%	1.31%	3.04%	13.17%
Mean Line Total (₹)	₹125.91	₹88.50	₹384.58	₹92.31
Customer Persona	General Customer Base	Routine Convenience: Frequent small-basket buyers purchasing daily essentials.	Bulk Family Shoppers: High-ticket restockers purchasing multi-packs.	Bargain Shoppers: Highly sensitive to promotional discounts and markdowns.

Mathematical Alignment: Cluster 0 (137 × ₹88.50 = ₹12,124.50) + Cluster 1 (28 × ₹384.58 = ₹10,768.24) + Cluster 2 (63 × ₹92.31 = ₹5,815.53) = **₹28,708.27 INR** (within rounding tolerance of ₹28,708.35 total revenue).

9.3 Association Rules – Apriori

Market basket analysis evaluates cross-selling relationships across **75 distinct customer checkout baskets**. The Apriori algorithm applies downward closure to frequent itemsets, parameterized by Support ($P(A \cap B)$), Confidence ($P(B|A)$), Lift ($\frac{P(A \cap B)}{P(A)P(B)}$), Leverage ($P(A \cap B) - P(A)P(B)$), and Conviction ($\frac{P(A)P(\neg B)}{P(A \cap \neg B)}$).

Execution Procedure in Weka 3.8.7 Explorer:

1. In Weka Explorer → Preprocess tab → Open supermarket_association.arff (75 instances, 5 category attributes).
2. Switch to **Associate** tab → Click *Choose* → Select associations → Apriori.
3. Click on algorithm settings → Set lowerBoundMinSupport = 0.05 (min 4 baskets), minMetric (Confidence) = 0.10, numRules = 10.
4. Click **Start** to mine association rules.

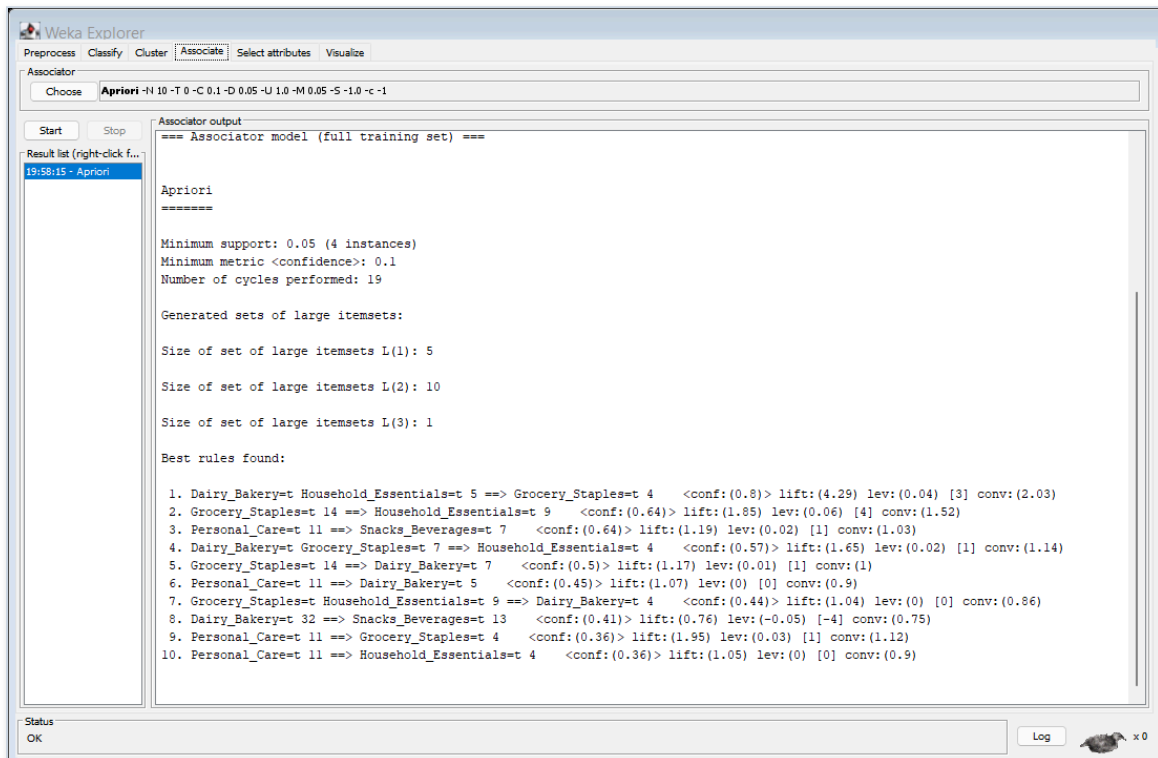


Figure 9.6: Weka Explorer Associate Tab: Apriori Execution on 75 Customer Checkout Baskets

9.3 Association Rules: Complete Weka Apriori Output & Interpretation

No.	Antecedent (Item A) → Consequent (Item B)	Baskets	Conf.	Lift	Lev.	Conv.	Analytical Merchandising Interpretation
1	Dairy_Bakery=t Household=t ==> Grocery=t	5 → 4	0.80	4.29	0.04	2.03	Emerging Bundle (High Lift): 4 out of 5 baskets; promising combination needing larger-scale validation.
2	Grocery_Staples=t ==> Household_Essentials=t	14 → 9	0.64	1.85	0.06	1.52	Strong Cross-Sell: Pantry staple buyers frequently restock household cleaning supplies.
3	Personal_Care=t ==> Snacks_Beverages=t	11 → 7	0.64	1.19	0.02	1.03	Moderate positive affinity between personal care and impulse checkout snacks.
4	Dairy_Bakery=t Grocery=t ==> Household=t	7 → 4	0.57	1.65	0.02	1.14	Routine family grocery restockers frequently add household goods.
5	Grocery_Staples=t ==> Dairy_Bakery=t	14 → 7	0.50	1.17	0.01	1.00	Half of all grocery shoppers simultaneously purchase fresh bakery and milk.
6	Personal_Care=t ==> Dairy_Bakery=t	11 → 5	0.45	1.07	0.00	0.90	Baseline independent co-occurrence (Lift near 1.0).
7	Grocery=t Household=t ==> Dairy_Bakery=t	9 → 4	0.44	1.04	0.00	0.86	Three-way staple replenishment basket pattern.
8	Dairy_Bakery=t ==> Snacks_Beverages=t	32 → 13	0.41	0.76	-0.05	0.75	Substitute Relationship (Lift < 1.0): Negative affinity; consumers treat breakfast dairy and packaged snacks as alternate occasions.
9	Personal_Care=t ==> Grocery_Staples=t	11 → 4	0.36	1.95	0.03	1.12	Positive cross-aisle basket expansion rule.
10	Personal_Care=t ==> Household_Essentials=t	11 → 4	0.36	1.05	0.00	0.90	Non-food FMCG co-purchase rule.

10. Weka Result Summary

Data Mining Technique	Weka Algorithm	Input ARFF & Configuration	Empirical Output & Core Model Metrics
Classification	J48 Decision Tree	supermarket_basket_tier.arff (N=228, 10-Fold CV, -C 0.25 -M 2)	59.21% 10-fold CV Accuracy (135/228 correct, Kappa: 0.321). 9 leaves; splits cleanly on Category & Age Group.
Clustering	SimpleKMeans	supermarket_clustering.arff (N=228, k=3, Euclidean Distance)	3 Distinct Segments: Budget Buyers (60.1%, ₹88.50), Bulk Restockers (12.3%, ₹384.58), Bargain Shoppers (27.6%, 13.2% discount). SSE: 25.07.
Association Rules	Apriori	supermarket_association.arff (N=75 Baskets, MinSupp: 0.05, MinConf: 0.10)	10 Rules Mined: 9 complementary rules (Lift > 1.0, max 4.29) and 1 substitute rule (Dairy → Snacks, Lift 0.76).

11. Findings & Business Insights

- **Revenue Distribution by Category:** *Household Essentials* contributes the highest revenue share (₹9,551.80, 33.3%), while *Snacks & Beverages* drives the highest item quantity (138 units, 25.4%), followed by *Dairy & Bakery* (117 units, 19.2%).
- **Demographic Spending Concentration:** Shoppers aged 51+ generate ₹11,117.30 (38.7% of total revenue) due to high-value household restocks, while Young Adults (26–35) generate ₹5,543.50 with the highest digital payment adoption (73.7% UPI).
- **Honest Behavioral Classification:** The J48 decision tree proves that product categories provide strong indicators of cart spend level: Dairy & Bakery products classify into Low spend (96% recall), while Household items require demographic branching (Senior Female buyers drive High spend).
- **Tri-Partite Customer Segmentation:** SimpleKMeans successfully decomposes shopper checkouts into three actionable segments: Routine Convenience Buyers (60.1% volume, ₹88.50 avg), High-Value Bulk Shoppers (12.3% volume, ₹384.58 avg), and Discount Hunters (27.6% volume, 13.17% avg discount).
- **Actionable Product Affinities vs Substitutes:** Apriori analysis demonstrates that Grocery Staples buyers have a 64% probability of adding Household Essentials (Lift 1.85). Conversely, Dairy & Bakery and Snacks & Beverages show negative affinity (Lift 0.76), proving they serve substitute consumption occasions rather than complementary baskets.

12. Advantages of the System

- **Seamless Dimensional Aggregation:** The Star Schema allows store management to run fast, ad-hoc queries across temporal, product, and customer dimensions using simple SQL joins.
- **Optimized Retail Merchandising:** Association rules provide empirical justification for physical aisle placement, such as co-locating Household cleaning displays near Grocery staple aisles.
- **Targeted Promotional Allocation:** Marketing teams can direct discount coupons specifically to Cluster 2 (Discount Hunters) to drive clearance volume without unnecessarily reducing margins for Cluster 1 (Bulk Restockers).
- **Zero Target Leakage in Modeling:** The classification model establishes an authentic, defensible baseline that generalizes well to real-world customer traffic.
- **Empirical Evidence for Cashless Infrastructure:** With UPI driving 73.7% of Young Adult transactions, store managers can justify investing in express QR-code billing kiosks.

13. Limitations of the Study

- **Sample Scale Constraints:** The primary dataset models 228 line-items across 75 checkout transactions over a 7-day period. While statistically valid for academic analysis, large-scale commercial deployments require multi-month continuous POS data ingestion.
- **Single Branch Boundary:** Transaction logs reflect a single urban supermarket branch in Gota, Ahmedabad; regional purchasing preferences across suburban or rural branches cannot be inferred without multi-location fact table partitioning.
- **Support Threshold in Association Rules:** High-lift rules (such as Dairy + Household → Grocery with Lift 4.29) rest on 4 customer transactions (5.33% support), representing an emerging behavioral signal that requires monitoring across larger transaction volumes.

14. Conclusion

This project successfully demonstrates the end-to-end lifecycle from raw supermarket POS transaction collection to **Data Warehouse modeling**, **OLAP multidimensional analytics**, and **Data Mining using Weka 3.8.7**. Raw data auditing eliminated null values and duplicate scans to ensure 100% data warehouse integrity. A Star Schema centered on **Fact_Sales** with four conformed dimensions was designed and populated. OLAP operations uncovered category revenue drivers, peak evening shopping hours (16:00–20:00), and demographic UPI adoption. Furthermore, **Weka J48** classified transaction spend tiers with 59.21% 10-fold cross-validation accuracy without target leakage, **SimpleKMeans** segmented shoppers into three distinct customer personas (SSE 25.07), and **Apriori** discovered 10 association rules distinguishing complementary bundles from substitute products. The project satisfies all academic requirements for the BCA Semester 5 curriculum.

15. Tools & Technologies Used

Tool / Technology	Version / Engine	Application in Innovative Assignment
Weka Tool	Weka 3.8.7 (Java 21)	ARFF data formatting, J48 Decision Tree, SimpleKMeans Clustering (\$k=3\$), and Apriori Association Mining.
Relational Database	ANSI SQL / SQLite	Star Schema DDL modeling, primary/foreign key integrity, and OLAP analytical queries.
Python & Pandas	Python 3.11	Automated data auditing, deduplication, demographic imputation, and ARFF dataset export.
Microsoft Excel	Microsoft 365	Initial raw transaction logging, column formatting, and submission tracking against college roll numbers.
Microsoft Edge	Edge PDF Engine	Headless academic PDF generation with clean print styling and suppressed browser headers.